

ME 7751 Homework 5

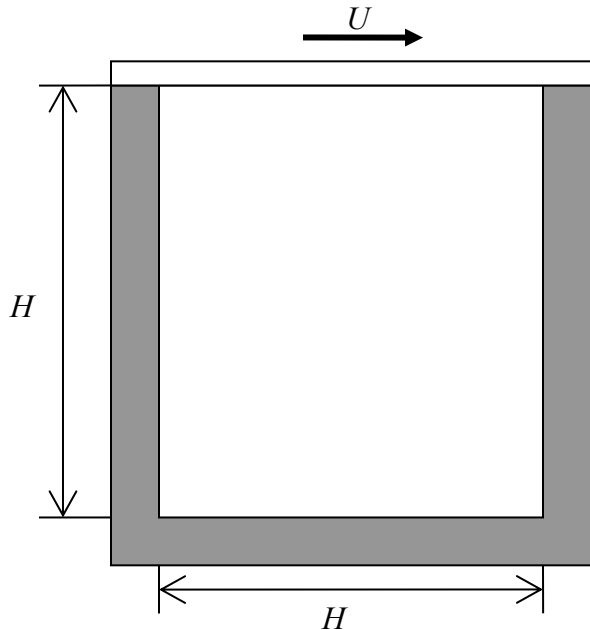
See homework submission guidelines on Canvas.

Problem 1

A two-dimensional square cavity with side H is filled with an incompressible Newtonian fluid with viscosity ν . The fluid is driven by the lid moving with a constant velocity U . The flow is characterized by a Reynolds number $Re = UH/\nu$. This problem is widely used as a benchmark to validate CFD models due to its simple geometry but nontrivial flow solution.

In this assignment you will numerically solve the cavity flow using an explicit projection method.

1. Describe the essential steps for the explicit projection method. Use finite differences on a rectangular staggered mesh for discretization and draw a schematic indicating the location of the three unknowns (pressure and two velocity components). Detail approximations made to estimate various derivatives. Detail how the divergence free constraint is enforced. Detail boundary conditions. Present the final discretized scheme in non-dimensional form.
2. Compute steady state solutions for $Re = 100$ and $Re = 400$. Use Gauss-Seidel SOR to solve the pressure equation to within an appropriate residual. Plot the steady state streamlines. Evaluate the divergence of the velocity field. Plot the x component of velocity along the vertical centerline. Plot the y component of velocity along the horizontal centerline. Compare your results to those found in the literature.¹
3. For $Re = 100$ and $Re = 400$, qualitatively show the centerline velocity profiles are converging to the literature values by considering three different grids with the grid resolution doubled each time.
4. Numerically compute the spatial convergence rate of your scheme. Detail how you computed the spatial convergence rate. Does it match what is expected based on your discretization?
5. Discuss method stability and stability criteria. Test the method stability numerically.



¹ Ghia U.K.N.G., Ghia K.N., and Shin C.T. High-Re solutions for incompressible flow using the Navier-Stokes equations and a multigrid method. Journal of computational physics, 1982, 48(3), 387-411.